Pantoprazole

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Continuing Education Activity

Pantoprazole is a proton-pump inhibitor (PPI) approved by the FDA for the treatment and maintenance of erosive esophagitis associated with gastroesophageal reflux disease, as well as for pathological hypersecretory conditions such as Zollinger-Ellison syndrome. In both inpatient and outpatient settings, pantoprazole remains a common therapeutic agent due to a favorable safety profile and availability in both adult and pediatric formulations. Clinicians may also prescribe pantoprazole off-label for eradicating *Helicobacter pylori*, preventing NSAID-induced ulcers, and prophylaxis against stress ulcers in patients who are critically ill.

This educational activity presents an in-depth review of pantoprazole's indications, dosing strategies, administration routes, contraindications, and necessary precautions. Special attention is given to the drug's mechanism of action, monitoring parameters, and potential adverse effects. Discussion of clinically significant drug-drug interactions reinforces the importance of careful prescribing in complex treatment regimens. The content supports appropriate use of pantoprazole across diverse clinical scenarios and emphasizes the collaborative role of the interprofessional team in optimizing outcomes.

Objectives:

- Identify the mechanism of action of pantoprazole.
- Assess the adverse effects associated with pantoprazole administration.
- Screen for contraindications against administering pantoprazole.
- Implement effective collaboration and communication among interprofessional team members to improve outcomes and treatment efficacy for patients who might benefit from pantoprazole therapy.

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Indications

Pantoprazole is a proton-pump inhibitor (PPI) widely used in hospitals and outpatient settings.

FDA-Approved Indications

Pantoprazole has been approved by the Food and Drug Administration (FDA) for the treatment of various disease processes, including therapy for erosive esophagitis associated with gastroesophageal reflux disease and the treatment of pathological hypersecretory conditions, including Zollinger-Ellison syndrome. This medication is also FDA-approved for managing erosive esophagitis. According to the American College of Gastroenterology, for patients presenting with typical symptoms of GERD (eg, heartburn and regurgitation) and lacking alarm features, an empirical course of once-daily PPIs, such as pantoprazole, administered 30 to 60 minutes before meals, is recommended for 8 weeks. For patients who experience symptom resolution, a trial discontinuation of the PPI may be appropriate. In cases where symptoms persist despite optimized treatment or recur upon cessation, esophagogastroduodenoscopy is indicated. Clinicians should also rule out cardiac causes.

Off-Label Uses

Pantoprazole has various off-label uses, including eradicating *Helicobacter pylori* bacteria and preventing peptic ulcer rebleeding and NSAID-induced ulcers. Critically ill patients may be given pantoprazole as stress ulcer prophylaxis. The Society of Critical Care Medicine and the American Society of Health-System Pharmacists endorse PPIs for stress ulcer prophylaxis. The administration of this medication is safe in adult and pediatric populations. The American College of Gastroenterology (ACG) 2025 guidelines support the use of PPIs, such as pantoprazole, in conjunction with other therapies for eosinophilic esophagitis. According to the ACG guidelines for upper GI bleeding, pre-endoscopic PPI therapy is not routinely recommended, as there is no evidence indicating a reduction in rebleeding or mortality. However, it may lower the prevalence of high-risk stigmata (eg, active bleeding, visible vessels) at endoscopy and reduce the need for endoscopic intervention. The selective administration of pantoprazole may be reasonable when endoscopy is delayed or unavailable, though supporting data are limited and of low quality. In contrast, post-endoscopic high-dose PPI therapy is strongly recommended for patients with ulcers demonstrating high-risk stigmata, as it significantly reduces rebleeding, surgical intervention, and mortality. Continued PPI therapy for 14 days post-intervention is advised to optimize outcomes.

Mechanism of Action

Proton pump inhibitors are categorized into 2 groups: the benzimidazole group and the imidazopyridine group. Pantoprazole falls in the benzimidazole group of PPIs. The primary difference between them is that benzimidazoles have a faster rate of metabolism, resulting in a shorter presence in plasma. Regarding the mechanism of action, pantoprazole irreversibly inhibits the H^+/K^+ ATP pumps. There is an increased rate of pantoprazole degradation at lower environmental pH levels. Therefore, it makes sense that this medication would work best in the stomach, where the H^+/K^+ ATP pumps are located (specifically within the parietal cells of the stomach lining). This is the final step in gastric acid production. Binding of pantoprazole to these pumps prevents acid secretion for up to 24 hours. After 24 hours, new pumps have been created, and thus a subsequent dose of pantoprazole is required to inhibit their action.

Pharmacokinetics

Absorption: Pantoprazole's onset of action is rapid, with the maximal effect occurring between 2 and 6 hours after administration. When standardized to omeprazole equivalents (OE = 1.00), pantoprazole has an OE of 0.23, lansoprazole has an OE of 0.90, esomeprazole has an OE of 1.60, and rabeprazole has an OE of 1.82.[2] This medication has a bioavailability of 77% when taken orally. The concomitant administration of antacids does not significantly affect the absorption of pantoprazole.

Distribution: The apparent volume of distribution is approximately 11 to 23.6 liters, with a primary distribution in extracellular fluid. Pantoprazole's serum protein binding is approximately 98%, predominantly to albumin.

Metabolism: Pantoprazole is metabolized in the liver, primarily through CYP2C19 demethylation and sulfation. Other metabolic pathways involve oxidation by CYP3A4.[11] The effect of metabolites is considered insignificant.

Excretion: Pantoprazole is eliminated primarily through the kidneys, with approximately 71% of the administered dose excreted in the urine and around 18% via the feces through biliary excretion; no unchanged drug is recovered in the urine.

Administration

Available Dosage Forms and Strengths

Pantoprazole is available in intravenous formulations, including 40 mg/100 mL, 40 mg/50 mL, and 80 mg/100 mL in 0.9% NaCl solutions and an injection powder for reconstitution in 40 mg per vial. Additionally, pantoprazole is available in oral forms, including delayed-release tablets in 20 mg and 40 mg strengths, and delayed-release granules for oral suspension in 40 mg per packet.

Adult Dosing

Correctly dosing pantoprazole requires a correct diagnosis, as different administration protocols are based on the condition being treated. There are several disease processes that pantoprazole helps manage. These include erosive esophagitis associated with gastroesophageal reflux disease, erosive esophagitis, and pathological hypersecretory conditions such as Zollinger-Ellison syndrome. For erosive esophagitis associated with gastroesophageal reflux disease, pantoprazole can be administered orally or intravenously. For oral administration, treatment is typically 40 mg daily or 20 mg daily for milder cases, for a total of 8 weeks. There is an optional "maintenance" period during which the same dose can continue for up to 12 months. Intravenously, 40 mg of pantoprazole is administered daily for 7 to 10 days. For Zollinger-Ellison syndrome, pantoprazole administration can also be oral or intravenous. For oral administration, treatment is typically 40 mg

twice daily. There is an option of titrating up to 240 mg if needed. Intravenously, the recommended dose is 80 mg of pantoprazole administered every 12 hours. As mentioned, pantoprazole also has various off-label uses.

For the eradication of *Helicobacter pylori* bacterial infections, pantoprazole can be incorporated into drug regimens that include various types of antibiotics. For all of these regimens, the recommended dosing is 40 mg of pantoprazole twice daily. A loading dose of 80 mg, followed by an infusion of 8 mg per hour, is recommended for the prevention of peptic ulcer rebleeding. For NSAID-induced ulcers, the recommended dosing is 20 to 40 mg orally each day. Pantoprazole administration can be with or without food, but taking it 30 minutes before a meal is preferable. The tablet should be swallowed whole and should not be crushed or chewed.

Specific Patient Populations

Hepatic impairment: No dosage adjustment is required. Hepatotoxicity results in mild elevation in transaminase levels.

Renal impairment: No dosage adjustment is required. A recent study suggests a decline in GFR with pantoprazole use; additional research is needed.

Pregnancy considerations: In one study, the use of proton pump inhibitors (PPIs) during early pregnancy did not demonstrate a significant increase in the risk of congenital malformations. Sibling-controlled analyses further suggested that PPIs are unlikely to be major teratogens. Additionally, a large cohort study indicated that PPI use during the second and third trimesters was associated with a subtle increase in the risk of preeclampsia. However, no such association was found with first-trimester use. Clinicians should weigh the risks and benefits of PPI therapy on an individual basis to ensure the best outcomes for both the mother and the fetus.

Breastfeeding considerations: Pantoprazole has been detected in breast milk. Maternal doses of 40 mg of pantoprazole daily produce low levels in milk and are not expected to cause any adverse effects in breastfed infants.

Pediatric patients: Pantoprazole is indicated for the short-term treatment of erosive esophagitis associated with gastroesophageal reflux disease. Oral pantoprazole is approved for patients aged 5 and older. Children weighing 15 to 40 kg should receive 20 mg once daily, and those weighing more than 40 kg should receive 40 mg once daily. For intravenous use, pantoprazole is approved in patients aged 3 months and older. Infants weighing <12.5 kg should receive 0.8 mg/kg once daily, while those weighing more than 12.5 kg may receive 10 mg daily. Children aged 1 to 17 should receive 10 mg if they weigh less than 15 kg, 20 mg if they weigh between 15 and 40 kg, and 40 mg if they weigh 40 kg. Intravenous therapy should be discontinued once oral administration is tolerated. Evidence for the use of proton pump inhibitors for GERD in preterm infants is limited and inconclusive.

Older patients: The use of proton pump inhibitors, including pantoprazole, is associated with an increased risk of *Clostridioides difficile* infection, pneumonia, bone loss, and fractures in older adults. The American Geriatric Society Beers Criteria recommends avoiding scheduled use for more than 8 weeks unless necessary for high-risk patients, such as those using oral corticosteroids or chronic NSAIDs, as well as for conditions like erosive esophagitis, pathological hypersecretory disorders, or when maintenance treatment is required due to failure of drug discontinuation or inadequate response to H2-receptor antagonists.

Adverse Effects

Even though pantoprazole is a relatively benign medication, there is still a risk of adverse effects. The primary adverse effects of pantoprazole include diarrhea, headache, upper respiratory tract infection, and abdominal pain. Long-term complications of pantoprazole use include diarrhea attributed to *Clostridium difficile* or microscopic colitis, small-intestinal bacterial overgrowth, vitamin B12 deficiency, iron deficiency, calcium deficiency, magnesium deficiency, bone demineralization, interstitial nephritis, and diminished absorption of medications such as clopidogrel. There is a rare risk of subacute cutaneous lupus erythematosus associated with pantoprazole administration, and Health Canada acknowledged this risk in December 2017. In older adults specifically, treatment with pantoprazole for 8 weeks or longer is not recommended. The Beers Criteria classifies pantoprazole as an inappropriate medication for the older population due to an associated increased risk of *Clostridium difficile* infection and bone density loss. A recent study has reported that QTc prolongation in critically ill patients; additional research is required.

Drug-Drug Interactions

- **Antiretrovirals:**
    - Rilpivirine: Concomitant use is contraindicated due to decreased antiviral effect and risk of drug resistance.
    - Atazanavir: Monitor for potential reduced efficacy; consult dosing recommendations for adjustments.
    - Nelfinavir: Avoid concomitant use due to decreased efficacy.
- **Warfarin:**According to product labeling, altered INR and prothrombin time are possible. Regular monitoring of INR is required, and warfarin dose adjustments may be necessary. However, one study suggests conflicting results.[33]
- **Drugs dependent on gastric pH for absorption:** Pantoprazole can lower gastric acidity, reducing the absorption of drugs such as iron salts, erlotinib, dasatinib, nilotinib, mycophenolate mofetil, and ketoconazole/itraconazole. Monitoring and possible dose adjustments may be necessary to maintain the effectiveness.

- **Methotrexate:** Concomitant use, especially at high doses, may increase and prolong serum methotrexate levels, potentially causing toxicity.[34]
- **Incompatibility**: Midazolam IV is incompatible with Y-site administration of IV pantoprazole.

Drug-Laboratory Interactions

Tetrahydrocannabinol levels: There have been instances of false-positive results in urine screenings for tetrahydrocannabinol in patients taking proton pump inhibitors (PPIs), including pantoprazole. Clinicians should interpret these results with caution.

Neuroendocrine tumor diagnostics: Proton pump inhibitor–induced acid suppression can elevate serum chromogranin A (CgA) concentrations, potentially leading to false-positive interpretations during neuroendocrine tumor diagnostics. Discontinuation of pantoprazole for at least 14 days before CgA measurement is recommended; retesting should be considered if levels remain elevated. Consistent use of the same assay laboratory is advised for serial assessments due to inter-assay variability in the reference range.

Contraindications

Pantoprazole is contraindicated in patients with a history of hypersensitivity to the drug itself, components of the formulation, or other benzimidazole PPIs, including omeprazole, lansoprazole, rabeprazole, esomeprazole, or dexlansoprazole. Hypersensitivity reactions include, but are not limited to, anaphylactic shock, pulmonary effects such as bronchospasm, angioedema, and urticaria. If a hypersensitivity reaction were to occur following the administration of pantoprazole, the infusion should stop immediately. Furthermore, pantoprazole should not be taken with rilpivirine (which is a non-nucleoside reverse transcriptase inhibitor) as it may reduce the drug's serum concentration.

Warning and Precautions

Intersitial nephritis: According to the Kidney Disease: Improving Global Outcomes (KDIGO) guidelines, it is advised to exercise caution regarding long-term use of proton pump inhibitors (PPIs) in individuals with unexplained chronic kidney disease (CKD), as well as in light of case reports that link proton pump inhibitor use to acute kidney injury (AKI) and interstitial nephritis. However, KDIGO indicates that proton pump inhibitors (PPIs) are generally effective and safe, although they may occasionally be associated with interstitial nephritis. KDIGO advises evaluating bleeding risk and recommends the use of PPIs when prescribing antiplatelet or antithrombotic therapy, particularly when these therapies are used in combination.

Gastric malignancy: The symptomatic response to pantoprazole does not exclude the presence of gastric malignancy. Further diagnostic evaluation should be considered in patients with suboptimal response or early symptom relapse after treatment. In older patients, endoscopic assessment should also be considered.

Injection site reactions: Thrombophlebitis has been associated with the use of intravenous pantoprazole sodium. The infusion site should be monitored regularly, and the catheter should be removed if clinically indicated.

Potential for exacerbation of zinc deficiency: Pantoprazole IV contains edetate disodium (EDTA), a chelating agent for divalent metal ions, such as zinc. Zinc supplementation should be considered in patients predisposed to zinc deficiency. Caution is advised with concurrent intravenous administration of other EDTA-containing agents.

***Clostridioides difficile*-associated diarrhea:** Proton pump inhibitor (PPI) therapy, including pantoprazole, may be associated with an increased risk of *Clostridioides difficile*-associated diarrhea, particularly in hospitalized patients. This diagnosis should be considered in patients with unexplained or persistent diarrhea.

Bone fracture: Patients receiving high-dose or long-term PPI therapy and those with kidney disease are at an increased risk of osteoporosis-related fractures. Clinicians should administer the lowest effective dose and shortest duration necessary.

Severe cutaneous adverse reactions: Serious cutaneous adverse reactions, including erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis, drug reaction with eosinophilia and systemic symptoms (DRESS), and acute generalized exanthematous pustulosis (AGEP), have been reported with PPIs. Discontinue pantoprazole at the first signs of severe skin reactions or hypersensitivity, and consider further evaluation.

Cutaneous and systemic lupus erythematosus: Infrequent cases of cutaneous lupus erythematosus (CLE) and systemic lupus erythematosus (SLE) have been reported in association with PPI therapy, including pantoprazole. Most cases of CLE are subacute (SCLE) and present weeks to years after initiation. SLE is less frequent and typically milder. Symptoms usually resolve within 4 to 12 weeks following drug discontinuation, although serologic abnormalities may persist longer.

Hypomagnesemia and mineral metabolism: Prolonged PPI use (>12 months) has been associated with hypomagnesemia, which may be symptomatic or asymptomatic, and can result in tetany, seizures, or arrhythmias. Secondary hypocalcemia or hypokalemia may also occur. In patients receiving long-term therapy or concurrent medications that predispose them to electrolyte imbalance (eg, digoxin, diuretics), periodic monitoring of magnesium and calcium levels should be considered. Supplementation or discontinuation may be warranted.

Fundic gland polyps: The use of long-term PPIs, including pantoprazole, is associated with an increased risk of fundic gland polyps, particularly after one year of treatment. These lesions are typically asymptomatic and incidentally identified during endoscopy. Limit therapy duration to the minimum necessary for clinical management.

Monitoring

Patients who are prescribed pantoprazole require monitoring for signs and symptoms of gastroesophageal reflux disease and peptic ulcer disease. If symptoms of these conditions arise, an increase in dosage or a change in medication should merit consideration by the clinician. Pantoprazole can reduce the concentration or increase the absorption of other medications, such as CYP2C19 inducers, bisphosphonate derivatives, amphetamines, fluconazole, and methotrexate, among others. Therefore, the affected efficacy of these medications also requires vigilance, and appropriate changes should be made when necessary. Finally, pantoprazole sounds and looks similar to the antipsychotic aripiprazole. Healthcare professionals and patients should acknowledge this and exercise caution to avoid mistakes. If UGIB is suspected, large-bore IV access and resuscitation are critical. FOBT may be ordered if there is suspicion of occult GI bleed. Bone mineral density should be monitored for patients receiving prolonged pantoprazole therapy. Furthermore, monitoring for *Clostridium difficile*-associated diarrhea (CDAD) is critical. Baseline and follow-up comprehensive metabolic panels with electrolytes should be obtained if long-term treatment is planned to monitor serum magnesium and calcium levels.

Toxicity

Signs and Symptoms of Overdose

There have not been a significant number of pantoprazole overdoses that have resulted in serious medical consequences. Laboratory studies have shown that single doses ranging from 709 to 887 mg/kg caused death in animal subjects (mice, rats, and dogs). Acutely, evidence of hypoactivity, tremor, and ataxia was observed in these animals.

Management of Overdose

There is no specific antidote for an event such as this. Furthermore, pantoprazole is not removable from the body via hemodialysis. As a result, treatment for pantoprazole overdose focuses on symptomatic relief. Contact the poison control center (800-222-1222) for the most up-to-date recommendations.

Enhancing Healthcare Team Outcomes

Pantoprazole is often prescribed by clinicians, including advanced practice providers, internists, and gastroenterologists. While the drug is effective for peptic ulcer disease, there have been many reports of overdose. The pharmacist should also perform a complete medication reconciliation when starting pantoprazole. Healthcare professionals should avoid writing prescriptions for a prolonged period without an indication and educate patients on drug safety, including storing the medication in a locked cabinet away from the reach of children. PPI prescriptions require careful review at hospital discharge to determine their continued necessity. Medication reconciliation is crucial for ensuring the appropriate use of medications and preventing potential long-term risks.

All interprofessional healthcare team members must collaborate effectively when patients are prescribed pantoprazole. This collaboration involves the transparent sharing of information and the meticulous coordination of therapeutic interventions, ensuring that patients receive the highest standard of care while minimizing the occurrence of adverse events. An interprofessional team approach and open communication among clinicians, gastroenterologists, pharmacists, and nurses are crucial to decreasing potential adverse effects and improving patient outcomes related to pantoprazole therapy.

Review Questions

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